Project Title: Dropout Risk Detection

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- Batch: A1

- Course: ML Mini Project

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Problem Statement (Revised)

- Many students drop out due to low performance and lack of engagement.
- Institutes need a reliable way to **identify at-risk students early**.
- Existing systems lack real-time prediction and behavioral analysis.
- This study focuses on using **student data + machine learning** to classify students as At-Risk/Not At-Risk.
- Goal: build an accurate prediction model to support early intervention

Dataset Details

Dataset Source

- **Name:** *Predict Students' Dropout and Academic Success*
- **Source:** Kaggle / UCI Machine Learning Repository

Number of Samples

- **Rows (instances): 4,424 students**
- **Features: 36 input features + 1 target column**

Feature Description

The dataset includes **student demographic, academic, financial, family, and economic details**.

- **Demographic:** gender, age, nationality
- **Academic:** course, previous grades, semester performance
- **Financial:** fees status, scholarship, debtor
- **Family:** parents’ education & occupation
- **Economic:** unemployment, inflation, GDP

Sample Data Snapshot

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	Marital st	Applicatic	Applicatic	Course	Daytime/e	Previous c	Nacionali	Mother's c	Father's q	Mother's c	Father's o	Displaced	Education	Debtor	Tuition fee	Gender	Scholarsh	Age at enr	Internatio	Curricular
2	1	8	5	2	1	1	1	13	10	6	10	1	0	0	1	1	0	20	0	0
3	1	6	1	11	1	1	1	1	3	4	4	1	0	0	0	1	0	19	0	0
4	1	1	5	5	1	1	1	22	27	10	10	1	0	0	0	1	0	19	0	0
5	1	8	2	15	1	1	1	23	27	6	4	1	0	0	1	0	0	20	0	0
6	2	12	1	3	0	1	1	22	28	10	10	0	0	0	1	0	0	45	0	0
7	2	12	1	17	0	12	1	22	27	10	8	0	0	1	1	1	0	50	0	0
8	1	1	1	12	1	1	1	13	28	8	11	1	0	0	1	0	1	18	0	0
9	1	9	4	11	1	1	1	22	27	10	10	1	0	0	0	1	0	22	0	0
10	1	1	3	10	1	1	15	1	1	10	10	0	0	0	1	0	1	21	1	0
11	1	1	1	10	1	1	1	1	14	5	8	1	0	1	0	0	0	18	0	0
12	1	1	1	14	1	1	1	23	14	6	8	1	0	0	1	0	0	18	0	0
13	1	1	1	12	1	1	1	13	28	10	10	1	0	0	1	0	1	18	0	0
14	1	1	2	16	1	1	1	13	27	5	10	1	0	0	1	0	0	19	0	0
15	1	17	1	11	1	16	1	1	1	5	8	1	0	0	1	0	1	21	0	0
16	1	1	1	6	1	1	1	23	27	6	6	1	0	0	1	0	1	18	0	0
17	1	1	1	15	1	1	1	13	27	10	4	1	0	0	1	0	0	20	0	0
18	1	9	1	10	1	1	1	13	28	6	9	1	0	0	1	0	0	18	0	0
19	1	8	2	12	1	1	1	13	1	6	5	1	0	0	1	0	0	18	0	0
20	1	1	1	8	1	1	1	3	14	4	6	1	0	0	1	0	0	20	0	0
21	1	1	1	16	1	1	1	13	14	8	8	1	0	0	1	0	0	18	0	0
22	1	1	3	2	1	1	1	1	1	10	9	0	0	0	1	0	1	21	0	0
23	1	9	4	13	1	1	1	1	28	5	8	1	0	0	1	0	0	20	0	0
24	1	1	4	12	1	1	1	13	14	2	2	1	0	0	1	0	0	18	0	0
25	1	1	4	14	1	1	1	1	28	5	8	1	0	0	1	0	1	19	0	0
26	1	1	1	12	1	1	1	13	14	4	8	0	0	0	1	0	0	19	0	0
27	1	1	1	10	1	1	1	13	28	10	10	1	0	1	1	0	1	18	0	0

Data Understanding

Types of Data

- Numerical: grades, age, GDP, inflation, unemployment
- Categorical: gender, course, marital status, scholarship
- Dataset includes continuous, integer, and categorical values.

Missing Values

- No missing values in the final dataset
- Missing and anomalous records were already handled during preprocessing by the dataset creators.

Data Distribution Insights

- Class imbalance present
- More Graduate samples than Dropout
- Semester grades and approved credits strongly influence the target
- Academic features are the most predictive factors

Preprocessing Plan

Handling Missing Values

- No missing values
- Dataset is already cleaned and ready to use

Encoding Categorical Variables

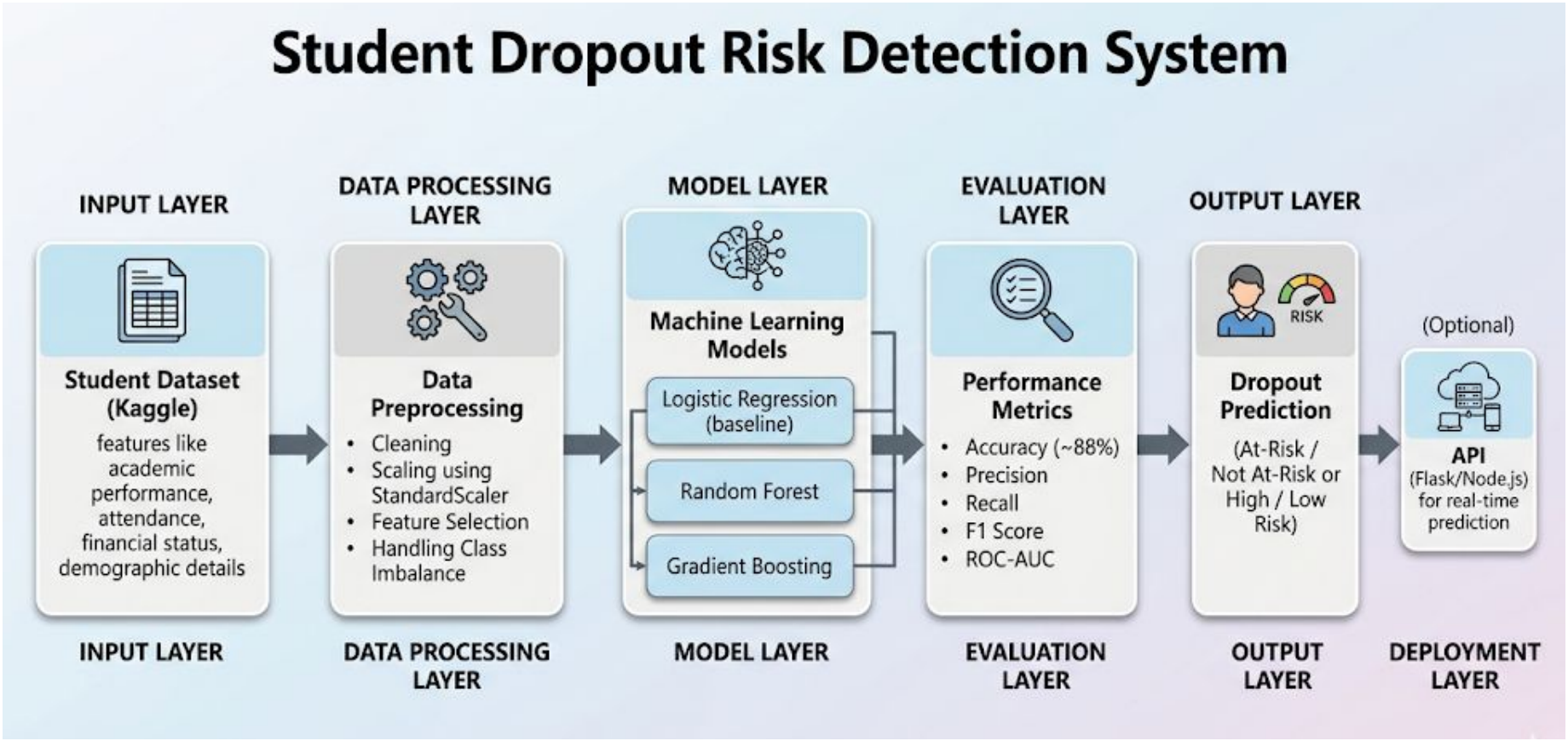
- No additional encoding required
- All categorical fields are already label encoded as numeric values
- Features can be directly used for ML models

Normalization / Scaling

- Use StandardScaler
- Important for Logistic Regression, SVM, DNN
- Less important for tree-based models

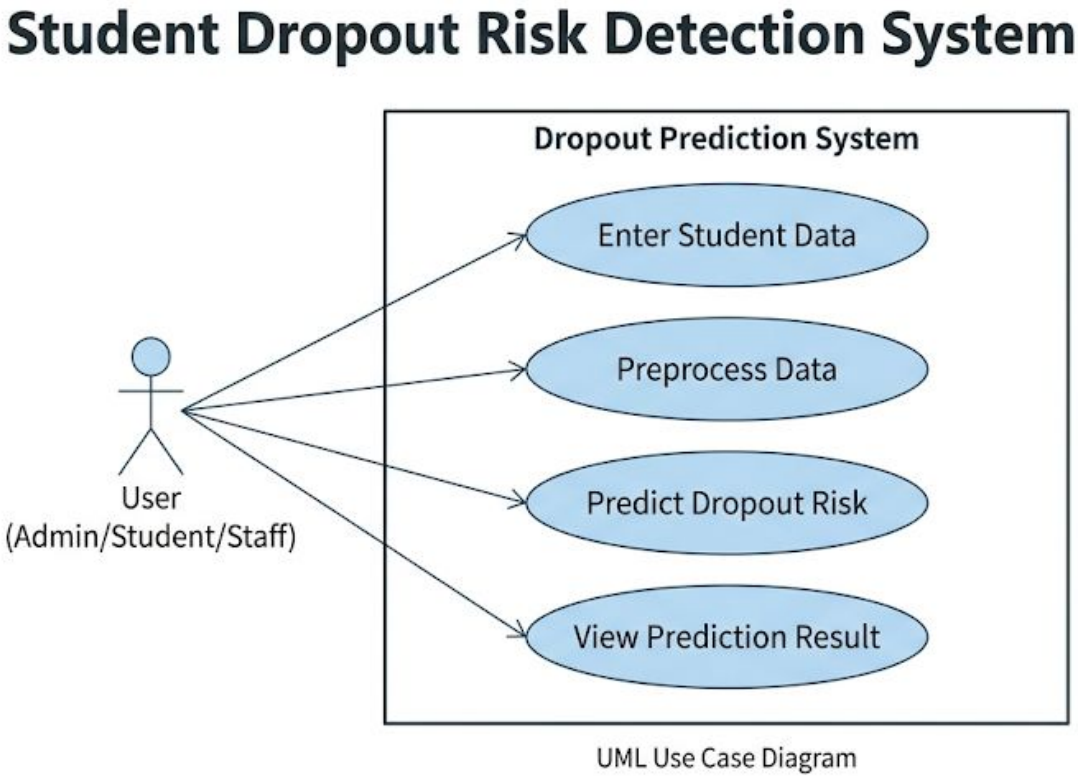
System Block Diagram

- Input → Processing → Model → Output



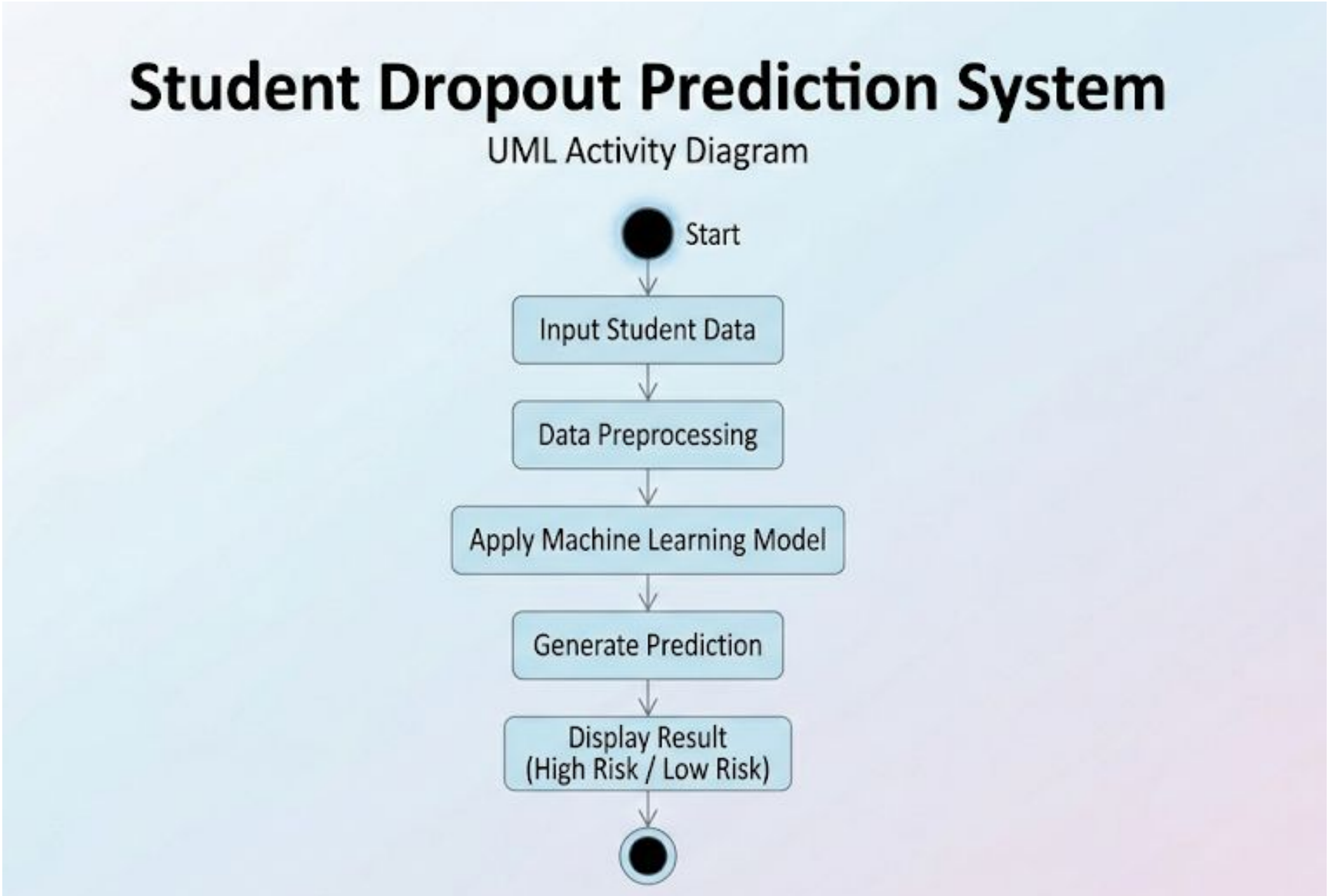
UML Diagrams

- Use Case Diagram



UML Use Case Diagram

- Activity Diagram



Algorithm Selection

Algorithms Shortlisted

- Logistic Regression
- Random Forest
- Gradient Boosting

Reason for Choosing Each

- Logistic Regression: simple, fast, and strong baseline classifier
- Random Forest: handles non-linearity, reduces overfitting, gives feature importance
- Gradient Boosting: high accuracy by learning from previous model errors

Baseline Model

- Baseline: Logistic Regression
- Chosen because it is easy to train, interpretable, and widely used for classification

Feature Selection

Selected Features

- **Personal:** age, gender, marital status, nationality, international
- **Academic background:** previous qualification, application mode/order
- **Semester performance:** enrolled, approved units, grades, evaluations
- **Financial/social:** fees status, debtor, scholarship, course
- **Economic/family:** unemployment, inflation, GDP, parents' education

Justification

- **Semester approved units + grades** → strongest academic predictors
- **Tuition fees, debtor, scholarship** → financial stress indicators
- **Age and family background** → personal support & responsibility factors
- **Economic indicators** → external stress on continuation

Domain Knowledge Used

- Low grades → poor academic progress
- Fee delays → financial hardship
- Scholarship → higher motivation
- Older/married students → more outside responsibilities

Evaluation Metrics for Project

Performance Metrics

- **Accuracy:** overall correct predictions (**~88%**)
- **Precision:** how many predicted dropouts were truly dropout
- **Recall:** how many real dropouts were correctly identified
- **F1 Score:** balance between precision and recall
- **ROC-AUC:** model's ability to separate dropout vs non-dropout

Why These Metrics?

- **Accuracy:** quick overall performance check
- **Precision:** reduces false alarms
- **Recall:** most important — avoid missing real at-risk students
- **F1 Score:** useful for imbalanced data
- **ROC-AUC:** best for comparing multiple models

Tools & Technologies

Programming Language

- **Python**
- Chosen for strong ML ecosystem, easy syntax, and fast experimentation

Libraries Used

- **NumPy** → numerical operations
- **Pandas** → data loading & preprocessing
- **Scikit-learn** → train-test split, models, metrics, scaling
- **Warnings** → suppress unnecessary warnings

Platform / Tools

- Google Colab
- Python environment with Scikit-learn
- Dataset from Kaggle

Initial Implementation

Data loaded successfully

```
======================================================================
        DROPOUT RISK DETECTION SYSTEM
        Real Student Dataset Analysis
======================================================================

📂 Dataset Loaded Successfully!
   Total Students  : 4424
   Total Features  : 34
   Missing Values  : 0
```

Preprocessing completed

```
============================================================
  STEP 3: DATA PREPROCESSING
============================================================

   Training samples : 3539
   Testing  samples : 885
   Features used    : 34
   Dropout in train : 1137 (32.1%)
   Dropout in test  : 284 (32.1%)
```

Logistic Regression model trained

```
============================================================
  STEP 4: MODEL TRAINING & EVALUATION
============================================================

------------------------------------------------------------
  🤖 Logistic Regression
------------------------------------------------------------
  Accuracy  : 88.02%
  ROC-AUC   : 0.9289

  Classification Report:
                precision    recall  f1-score   support

   Not Dropout       0.88      0.95      0.92       601
       Dropout       0.87      0.73      0.80       284

      accuracy                           0.88       885
     macro avg       0.88      0.84      0.86       885
  weighted avg       0.88      0.88      0.88       885

  Confusion Matrix:
                  Predicted No  Predicted Yes
    Actual No    :    TN= 571         FP=  30
    Actual Yes   :    FN=  76         TP= 208
```

Initial Results

- **Accuracy: 88.02%**
- **ROC-AUC: 0.9289**
- **Precision: 0.87**
- **Recall: 0.73**
- **F1-Score: 0.80**

```
Confusion Matrix:
                  Predicted No  Predicted Yes
  Actual No    :    TN= 571         FP=  30
  Actual Yes   :    FN=  76         TP= 208
```

Observations

- Strong overall accuracy (88%)
- High performance in identifying non-dropout students
- Recall of 73% means most at-risk students were detected
- Some 76 dropout students were missed, so recall can still improve
- Good baseline model for comparison

Challenges Faced

- Understanding dataset features and their meaning
- Handling class imbalance in dropout data
- Selecting the most relevant features
- Choosing appropriate ML models
- Improving recall for dropout prediction

Plan for Next Phase

Model Improvement

- Tune hyperparameters for better recall
- Try XGBoost / LightGBM / Deep Learning
- Use SMOTE for better class balance
- Perform advanced feature selection

UI / Deployment Plan

- Build a web dashboard using Streamlit / Flask
- Input student details manually or via CSV
- Show dropout risk score + intervention suggestions
- Deploy on Render / Hugging Face / Streamlit Cloud

Paper Writing Start

- Document problem statement, dataset, methodology, results
- Compare shortlisted models
- Include graphs, confusion matrix, ROC curves
- Write future scope and real-world education impact

Deliverables

Deliverable 1: Dataset file (.csv)

- Student dataset from Kaggle

Deliverable 2: Jupyter Notebook

- Data loading
- Data preprocessing
- At least one ML model
- Model evaluation

Deliverable 3: GitHub Repository

- Complete project uploaded on GitHub
- Link:

https://github.com/taanshubabariya-cyber/AIML-ML-LAB-03/tree/main/Mini_Project

Deliverable 4: Block Diagram Image

- System architecture diagram

References

- Kaggle Dataset: Predict Students' Dropout and Academic Success

- Villar & Andrade (2024) – Discover AI
- Marcolino et al. (2025) – Scientific Reports
- Cho et al. (2023) – Applied Sciences
- Kim et al. (2023) – Applied Sciences